(O2) compensation costs at $n = 2$: the single-cheap-feature game, two regimes, and what is left

fork O2

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Abstract

We set up and solve numerically the single-cheap-feature compensation game of (O2) at $n = 2$, for the programme maximisers at $b = 2$ ($s = \frac{3}{2}$) and $b = 2.5$ ($s = 1.2$), all primes $5 \le p \le 41$, every feature of the support, with all quadratic compensation channels of the note (own partner, riding, stray outputs), Lorentz costs, the sharp single-feature bonus (computed), single-character Riesz tests and the quadratic (L^2-over-T) test. Findings. (1) *Linear gain models fail.* Under any gain bound that is linear in $|\beta_{l,T}|$ near 0 (Theorem A's $\frac{1}{2}R$ or the conjectured sharp $c_1 R$) the network wins, for some feature at every sparse prime. We prove a dichotomy that explains this exactly: at a character $\chi_{l,T}$ the compensation can hide the sign-flipped mode to first order *for free* if and only if the channels supply at least $|B|$ distinct compensating vectors (Chebotarev). Partner and riding supply $2|S| - 1 < |B|$; stray outputs (only at $T = [n]$) supply up to $2|S|$ more, and then hiding is exact. In the dense case ($S = F$) the free-hiding slope is exactly $\frac{1}{2}$: Theorem A is exactly tight against compensation. (2) *The bonus is quadratic near the IF block*, with an explicit constant: for three atoms near the IF block, $\operatorname{Re}\beta - a = \frac{1-r}{8+r}\,|\beta_1 + \beta_2 + \overline{\beta_{12}}|^2/a + O(\varepsilon^3)$ (proved), i.e. $0.346\,R^2/a$ for $R = \max_T |\beta_T|$ and $0.0385\,R^2/a$ if all interference sits on $T = \{1, 2\}$ (at $b = 2$). The interior maximum of bonus$/R$ is reached on an explicit two-atom branch, $G_2(\rho) = 3\rho - \rho^3/c^2 - 1$. (3) *With the sharp (computed) bonus the prover wins in every case computed, but at $b = 2$ only narrowly with single-character tests.* The relevant quantity is the safety ratio $\sigma(p, l)$ (protection over bonus, minimised over the network's interference profile; $\sigma \ge 1$ iff $V \ge 0$). At $b = 2$: the task's game (partner + riding) has $\min \sigma = 1.015$ ($p = 17$, $l = 5$); with stray outputs and single-character Riesz tests $\min \sigma = 1.051$ with the true joint Lorentz account (0.959 if costs are charged per test); with the quadratic test $\min \sigma = 1.69$. At $b = 2.5$ all ratios are ≥ 2.35 (Riesz) and ≥ 4.1 (quadratic). The critical configurations are two-neuron cheap features (an explicit two-atom bonus branch) at $|\beta_{l,T}| \approx 0.5$–$0.8\,a_l$. Dedicated compensators (with sibling accounting) never help. Allocation moves *do* help the network at first order (the prover's best μ is not $\mu^\star$; forcing $\mu = \mu^\star$ loses, down to 0.655), but the coverage gap they open bounds the effect: the best moves found (local search on the tightest features) lower σ by $\le 27\%$ (Riesz, separate), 5–7% (joint), $< 1\%$ (quadratic test), and never below 1 in the joint or quadratic game in the cases searched. (4) The remaining gap for (O2) at $n = 2$ is precisely: a rigorous form of the sharp bonus function G^* (a 3-variable inequality on atom systems; only its linear envelope, Theorem A, is proved), and a p-uniform lower bound for the protection curves Π_T, which are explicit SOCP values attached to the support and binding set of each programme vertex.

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1 The game

Allocation and units. $n = 2$, $\nu = 3$, $s = 3/b \in [1, \frac{3}{2}]$, $r = 1/s$. We fix a programme maximiser $y^\star$ ($\sum_j y_j^{\star r} = 1$, support S, binding half-set $B_+ = \{\delta \in F : \sum_j y_j^\star \kappa_j(\delta) = \Gamma_s\}$, $B = B_+ \cup (-B_+)$), and work in units where $\sum_l Q_l = 1$. Then the interference-free (IF) amplitudes are $a_j = 3^{1-s} y_j^\star$ and the IF margin is $3^{1-s}\Gamma_s = \sum_j a_j K_j(\mu)$ for every probability μ on B, $K_j(\mu) = \sum_\delta \mu_\delta \kappa_j(\delta)$. Programme data were recomputed with the repository solvers `prog2`/`prog3` (best of both, plus exhaustive vertex enumeration `progx` for supports ≤ 4 at $p \le 23$); the KKT measure $\mu^\star$ on B was obtained by an LP and has zero residual in every case. At $b = 2$ the supports are sparse for $p \ge 11$ ($|S| = 2$–5, $|B_+| = |S|$); at $b = 2.5$ the support is all of F for $p \le 17$.

Characters, coefficients, channels. Fix a cheap feature $l \in S$ and $T \in \{\{1\}, \{2\}, \{1,2\}\}$, $\chi_T = 2l\mathbf{1}_T$. With the convention $\Phi_{-m} = \overline{\Phi_m}$, $c_m(\delta) = 1 - \rho^{-m\delta}$ (so $c_{-m} = \overline{c_m}$), the coefficient of $x \mapsto g(x, \delta)$ at χ_T is $\hat{g}_\delta(\chi_T) = \frac{1}{2}\sum_{m \in \mathbb{Z}_p^\times} \phi_m(\chi_T)\, c_m(\delta)$. The own sign-flipped mode is $m = l$, of modulus $R_T = |\beta_{l,T}|$. By the channel lemma of the note (Lemma *channels*) and the paid/unpaid classification (Section *Architecture*), the quadratically paid compensating modes are:

channel	vector m	payer (Lorentz cost on its clean amplitude)	characters
own partner	$-l$	l (input $3l$ on T)	all T
riding	$\pm j$, $j \in S \setminus \{l\}$	j (input $2l \mp j$ on T)	all T
stray output	$2l \pm q$, $q \in S$, $\neq l$	q (output m on feature-q neurons)	$T = [n]$ only
shared (F2)	$\pm j$ with $2l \equiv \pm j \pm q$, $j \neq q \in S$	free	all T
dedicated	any m	all factors unpaid: concave cost	all T

A payer q carrying modes z_m (all characters together) keeps at most $\sqrt{a_q^2 - \sum |z_m|^2}$ of clean amplitude (Lemma *lorentz* of the note and Aczél), which enters (MI) weighted by $K_q(\mu)$. Siblings of riding and stray modes are assumed cancelled for free (this favours the network; for stray outputs on an IF block they are genuinely absent). The own partner is charged like riding (as specified); the variant "partner free" is reported separately.

Gain. By the paper's Lemma 6 the bonus $\mathrm{Re}\,\beta_l - a_l$ is a function of one atom system. The *sharp bonus*

$$G^*(\rho_1, \rho_2, \rho_{12}) = \sup\{\, |A| - a :\ a = 3^{1-s} Q^s = 1,\ |\beta_T| \le \rho_T \,\}$$

(atoms $\beta_i \in \mathbb{C}$, $\xi_i \in \mathbb{T}^2$) is computed by SLSQP with analytic gradients, widths $N \le 4$ (spot checks with $N \le 6$), IF-block and random starts, and continuation; it is a *lower* bound for the true

supremum. By slot symmetry we use $\rho_1 = \rho_2 = \rho_s$. Linear models: Theorem A, $G \le \frac{1}{2}\max_T R_T$ (proved); conjectured sharp $c_1 = 1 - 3^{1-s}$.

Tests. (i) Single-character Riesz tests at each χ_T: $h \le \mathbb{E}g_\delta - |\hat{g}_\delta(\chi_T)|$. (ii) The quadratic test of the note (Theorem *quad*, remark (c)): detection $\sqrt{2}\, f\, \|(\hat{g}_\delta(\chi_T))_T\|_2$ with $f = f(\gamma_{\max}) = 0.672$ in general and $f = 1$ when the interference sits on one T. For (ii) we ignore pair-sum and 2χ modes (type (R)), i.e. we assume the relevant dissociation.

Protection and safety ratio. For a test T and $\mu \ge 0$ symmetric on B put

$$\mathrm{Prot}_T(R,\mu) = \min_{\text{modes}} \Big[\sum_\delta \mu_\delta |\hat{g}_\delta(\chi_T)| + \sum_{q\in S} K_q(\mu)\Big(a_q - \sqrt{a_q^2 - \textstyle\sum_{m\to q}|z_m|^2}\Big)\Big], \qquad \Pi_T(R) = \max_{K_l(\mu)=1} \mathrm{Prot}_T(R,\mu).$$

Costs of different tests are charged separately (a relaxation in favour of the network). The network beats IF with cheap feature l only if some profile (ρ_s, ρ_{12}) has $a_l\, G^*(\rho_s,\rho_{12}) > \max\big(\Pi_{\{1\}}(\rho_s a_l), \Pi_{[2]}(\rho_{12} a_l)\big)$. We report

$$\sigma(p,l) = \min_{(\rho_s,\rho_{12})\neq 0} \frac{\max\big(\Pi_{\{1\}}(\rho_s a_l),\, \Pi_{[2]}(\rho_{12} a_l)\big)}{a_l\, G^*(\rho_s,\rho_{12})},$$

so $\sigma \ge 1$ is exactly the statement $V \ge 0$ of the task (the task's $V = \max_\mu \min_{R,\dots}[\cdots]$ is ≤ 0 always, $= 0$ at $R \to 0$; the informative number is the ratio). The minimum is over the grid $\rho \in \{0, .02, .05, .1, .15, .2, .3, .4, \dots, 1.7, 3^{s-1}\}^2$.

2 Structure of the game (proved)

Throughout this section the allocation, l, T and the channel family $M_T \subseteq \mathbb{Z}_p^\times \setminus \{l\}$ (with payers) are fixed. Write $Y_m(u) = \frac{1}{2}\sum_{\delta\in B} u_\delta \overline{c_m(\delta)}$ for $u \in \mathbb{C}^B$.

Lemma 1 (dual form, convexity, slope)**.**

$$\Pi_T(R) = \max\Big\{ R\,\mathrm{Re}\, Y_l(u) - \sum_q a_q\Big(\big\|\big(K_q(\mu), (Y_m(u))_{m\to q}\big)\big\|_2 - K_q(\mu)\Big) : \\ |u_\delta| \le \mu_\delta,\ \mu \textit{ symmetric},\ K_l(\mu) = 1,\ Y_m(u) = 0\ (m \textit{ free})\Big\}.$$

Consequently Π_T *is convex on* $[0,\infty)$ *with* $\Pi_T(0) = 0$, $\Pi_T(R)/R$ *is non-decreasing, and*

$$\lim_{R\downarrow 0} \Pi_T(R)/R = h_T := \max\{\mathrm{Re}\, Y_l(u) :\ |u_\delta| \le \mu_\delta,\ K_l(\mu) = 1,\ Y_m(u) = 0\ \forall m \in M_T\}.$$

Proof. $\mu_\delta|w| = \max_{|u_\delta|\le\mu_\delta} \mathrm{Re}(\bar{u}_\delta w)$, and for $f(z) = a - \sqrt{a^2 - \|z\|^2}$ ($\|z\| \le a$) the conjugate of Kf is $Kf^*(y/K) = a(\sqrt{K^2 + \|y\|^2} - K)$. The objective of Prot_T is convex in the modes and linear in (μ, u), the (μ,u)-set is compact ($\kappa_l > 0$ on B bounds μ), so Sion's theorem exchanges min and max; the own mode contributes $\mathrm{Re}\sum \bar{u}_\delta \frac{R}{2} c_l(\delta) = R\,\mathrm{Re}\, Y_l(u)$. Π_T is a maximum of affine functions of R with value ≤ 0 at $R = 0$ and ≥ 0 (take $u = 0$), so it is convex with $\Pi_T(0) = 0$, hence $\Pi_T(R)/R$ is non-decreasing. As $R \to 0$, $\Pi_T(R)/R = \max[\mathrm{Re}\, Y_l - \mathrm{cost}/R]$; the costs are ≥ 0 and vanish exactly when all $Y_m = 0$, so by compactness the limit is h_T. □

Corollary 2 (linear gain models)**.** *If the bonus is bounded by* $g\max_T R_T$ *(e.g.* $g = \frac{1}{2}$*, Theorem A, or* $g = c_1$*), then in the relaxed game the prover wins against every profile iff* $g \le \min_T h_T$*. If* $g > \min_T h_T$ *the network wins with arbitrarily small* R *on the minimising* T*.*

Proof. $\Pi_T(R) \geq h_T R$ by Lemma 1; conversely $\Pi_T(R) = h_T R + o(R)$. □

Proposition 3 (free hiding: a Chebotarev dichotomy)**.** *Let M_T consist of distinct residues $\neq 0, l$ (each with a free complex coefficient). Then $h_T = 0$ (exact first-order hiding) if and only if $|M_T| \geq |B|$. In particular:*

(a) with partner and riding only, $|M_T| = 2|S| - 1 < 2|S| \leq |B|$ at every vertex, so $h_T > 0$ for every T (this is the quantitative content of Proposition hide*);*

(b) at $T = [n] = \{1, 2\}$ the stray outputs add the residues $2l \pm q$ ($q \in S$). Whenever $|M_{[2]}| \geq |B|$, which holds for every sparse support computed ($p = 11, \ldots, 41$ at $b = 2$; $p \geq 19$ at $b = 2.5$), $h_{[2]} = 0$: the sign-flipped mode at $\chi_{l,[2]}$ can be hidden from every binding direction at quadratic cost.

Proof. $h_T = 0$ iff $c_l|_B \in \operatorname{span}\{c_m|_B : m \in M_T\}$ (the dual SOCP has a feasible u with $\operatorname{Re} Y_l(u) > 0$ iff some u annihilates the span but not c_l; scale it into $|u| \leq \mu$). Writing $e_k(\delta) = \rho^{-k\delta}$, $c_l = \sum_m z_m c_m$ on B means $\sum_{k \in K} w_k e_k = 0$ on B with $K = M_T \cup \{0, l\}$, $w_l = 1$, $w_m = -z_m$, $w_0 = \sum z_m - 1$, i.e. a relation with $\sum_k w_k = 0$, i.e. a relation among the e_k on $B \cup \{0\}$ with $w_l \neq 0$. By Chebotarev's theorem every square minor of the Fourier matrix of $\mathbb{Z}_p$ is non-zero, so any $\min(|K|, |B| + 1)$ of the e_k are independent on $B \cup \{0\}$. If $|K| \leq |B| + 1$ there is no non-trivial relation at all. If $|K| \geq |B| + 2$ the relations on $B \cup \{0\}$ form a space of dimension $|K| - |B| - 1$, and those with $w_l = 0$ (relations among $K \setminus \{l\}$) a space of dimension $|K| - |B| - 2$; so a relation with $w_l = 1$ exists. $|K| \geq |B| + 2$ iff $|M_T| \geq |B|$. For (a), every vertex has $|B_+| \geq |S|$. □

Proposition 4 (dense case: Theorem A is exactly tight against compensation)**.** *If $S = F$ and $B = \mathbb{Z}_p^\times$ (e.g. $p = 5, 7$ at $b = 2$, $p \leq 17$ at $b = 2.5$), then $h_T = \frac{1}{2}$ for every T and every channel family containing partner and riding.*

Proof. Then $M_T \supseteq \mathbb{Z}_p^\times \setminus \{l\}$. Put $U(k) = \sum_{\delta \neq 0} u_\delta \rho^{k\delta}$; the constraints read $U(m) = U(0)$ for $m \neq 0, l$. Extend u to $\mathbb{Z}_p$ by $u_0 = -U(0)$; its transform $\hat{u}(k) = u_0 + U(k)$ vanishes off $\{l\}$, so $u_\delta = \frac{1}{p}\hat{u}(l)\rho^{-l\delta}$ has constant modulus $|\hat{u}(l)|/p$. Then $\operatorname{Re} Y_l = -\frac{1}{2} \operatorname{Re} \hat{u}(l)$, and $K_l(\mu) \geq \frac{|\hat{u}(l)|}{p} \sum_{\delta \neq 0} \kappa_l(\delta) = |\hat{u}(l)|$ since $\sum_{\delta \neq 0} \kappa_l(\delta) = p$. So $h_T \leq \frac{1}{2}$, with equality for $\hat{u}(l) = -1$, $\mu_\delta = 1/p$. □

So with the proved gain constant $\frac{1}{2}$ the dense game has value exactly 0 (approached as $R \to 0$), and with any constant $< \frac{1}{2}$ (e.g. c_1) the prover wins with ratio $\frac{1}{2c_1} = 1.183$ at $b = 2$; with costs $\Pi_T(R)/R$ rises above $\frac{1}{2}$ for $R > 0$ (e.g. $\Pi/R = 0.512$ at $R = 0.1a_l$, 0.61 at $R = a_l$, $p = 5$, $b = 2$).

Proposition 5 (second-order bonus at the IF block, $n = 2$)**.** *Let three atoms $\beta_i = \frac{1}{3}(1 + x_i)e^{iy_i}$, $\xi_i = (\omega^i e^{i\theta_i}, \omega^i e^{i\phi_i})$ ($\omega = e^{2\pi i/3}$, $i \in \mathbb{Z}_3$) be within ε of the IF block, and $a = 3^{1-s}Q^s$. Then, with $\Sigma_\beta := \beta_1 + \beta_2 + \overline{\beta_{12}}$,*

$$|A| - a = \tfrac{1}{2}\big[(1 - r)\operatorname{Var}(x) - \operatorname{Var}(y)\big] + O(\varepsilon^3) \;\leq\; \frac{1 - r}{8 + r}\,|\Sigma_\beta|^2 + O(\varepsilon^3),$$

and the constant is attained. Hence near the IF block (whenever the bonus is ≥ 0, so that $\varepsilon \lesssim \max_T |\beta_T|$)

$$\textit{bonus} \leq \frac{9(1 - r)}{8 + r}\,\frac{R^2}{a} \;\; (R = \max_T |\beta_T|), \qquad \textit{bonus} \leq \frac{1 - r}{8 + r}\frac{|\beta_{12}|^2}{a} \;\; \textit{if } \beta_1 = \beta_2 = 0, \qquad \frac{\textit{bonus} \cdot a}{\sum_T |\beta_T|^2} \leq \frac{3(1 - r)}{8 + r},$$

up to $O(R^3)$: constants 0.346, 0.0385, 0.115 *at $b = 2$ and* 0.170, 0.0189, 0.0566 *at $b = 2.5$.*

Proof. $|A| = 1+\bar{x}-\frac{1}{2}\operatorname{Var} y+O(\varepsilon^3)$ and $a = 3^{1-s}\big(3^{-r}\sum(1+x_i)^r\big)^s = 1+\bar{x}-\frac{1}{2}(1-r)\operatorname{Var} x+O(\varepsilon^3)$ (use $s(s-1)r^2 = 1-r$). For real z on $\mathbb{Z}_3$, $\operatorname{Var} z = 2|\hat{z}|^2$ with $\hat{z} = \frac{1}{3}\sum z_i\omega^i$. Linearising, $\beta_1 = \hat{x}+i\hat{y}+i\hat{\theta}$, $\beta_2 = \hat{x}+i\hat{y}+i\hat{\phi}$, $\overline{\beta_{12}} = \hat{x}-i\hat{y}-i\hat{\theta}-i\hat{\phi}$ up to $O(\varepsilon^2)$, so $\Sigma_\beta = 3\hat{x}+i\hat{y}+O(\varepsilon^2)$: the slot phases cancel. Hence the bonus is $\frac{1-r}{9}|\Sigma_\beta - w|^2 - |w|^2 + O(\varepsilon^3)$ with $w = i\hat{y}$, whose maximum over w is $\frac{c}{1-c}|\Sigma_\beta|^2$, $c = \frac{1-r}{9}$, i.e. $\frac{1-r}{8+r}|\Sigma_\beta|^2$. It is attained, because $\hat{\theta}, \hat{\phi}$ then place β_1, β_2 freely. If the bonus is ≥ 0 then $|\hat{y}| \lesssim |\hat{x}| \lesssim |\Sigma_\beta|$, and $\hat{\theta}, \hat{\phi}$ are controlled by β_1, β_2, so $\varepsilon \lesssim \max_T |\beta_T|$. The three displayed constants follow from $|\Sigma_\beta| \leq 3R$, $|\Sigma_\beta| = |\beta_{12}|$, $|\Sigma_\beta|^2 \leq 3\sum|\beta_T|^2$. □

Numerically (Monte Carlo over 2000 perturbations of size 10^{-3}) the sup of bonus$\cdot a/|\Sigma_\beta|^2$ is 0.038467 vs. $\frac{1-r}{8+r} = 0.038462$ ($b = 2$) and 0.018837 vs. 0.018868 ($b = 2.5$); the global searches give $G^*_{\mathrm{sym}}(\rho)/\rho^2 \to 0.346$ and $G^*(0,\rho)/\rho^2 \to 0.0385$. This confirms the main thread's question: *near IF the bonus is quadratic*, with $\sup \text{bonus}\cdot|A|/\sum_T|\beta_T|^2 \to \frac{3(1-r)}{8+r}$. Proposition 5 is local (three atoms, ε small); away from the IF block the bonus is larger relative to R^2:

Proposition 6 (the two-atom branch)**.** *Two atoms $c\,e^{\pm i\varphi/2}$ with $c = \frac{1}{2}(\frac{3}{2})^{s-1}$ (so $a = 1$), $\xi_1 = (1,1)$, $\xi_2 = e^{i\varphi}(e^{iu}, e^{iu})$ with $u = \pi - 2\arcsin(\rho/2c)$ and $\varphi/2 = 3\arcsin(\rho/2c) - \frac{\pi}{2}$ have $|\beta_1| = |\beta_2| = |\beta_{12}| = \rho$ and bonus*

$$G_2(\rho) = 3\rho - \rho^3/c^2 - 1 \quad (\rho \leq c), \qquad G_2(\rho) = 2c - 1 = (\tfrac{3}{2})^{s-1} - 1 \quad (\rho \geq c).$$

$G_2(\rho)/\rho$ is maximal at $\rho_* = (c^2/2)^{1/3}$: $\rho_* = 0.572$, $G_2/\rho = 0.379$ at $b = 2$. For $\rho \in [0.47, 0.61]$ ($b = 2$) this branch is the computed G^*_{sym}; it is the configuration that makes the original game tightest (Section 3).

Proof. $|\beta_T| = 2c|\cos(\cdot)|$ with arguments $u/2$, $u/2$, $u+\varphi/2$; with $\sin(\varepsilon/2) = x := \rho/2c$, $u = \pi-\varepsilon$ and $u+\varphi/2 = \pi - \arccos x$ one gets $\varphi/2 = 2\arcsin x - \arccos x = 3\arcsin x - \frac{\pi}{2}$ and $A = 2c\cos(\varphi/2) = 2c\sin(3\arcsin x) = 2c(3x-4x^3) = 3\rho - \rho^3/c^2$. □

3 Numerical solution of the game

3.1 Programme data

p	$\Gamma_{3/2}(p)$	S	B_+	KKT weights $\lambda = 2\mu^\star$ on B_+
5	0.88388	$\{1,2\}$	$\{1,2\}$	0.500, 0.500
7	0.67358	$\{1,2,3\}$	$\{1,2,3\}$	0.333, 0.333, 0.333
11	0.53573	$\{2,5\}$	$\{2,5\}$	0.440, 0.560
13	0.49777	$\{1,3,5\}$	$\{1,3,5\}$	0.391, 0.348, 0.261
17	0.44834	$\{3,4,5\}$	$\{4,5,6\}$	0.493, 0.113, 0.393
19	0.44215	$\{6,8,9\}$	$\{2,6,9\}$	0.256, 0.124, 0.619
23	0.40293	$\{4,5,6,8\}$	$\{4,5,9,11\}$	0.213, 0.056, 0.353, 0.378
29	0.38992	$\{1,4,10,12\}$	$\{3,6,7,14\}$	0.350, 0.150, 0.350, 0.150
31	0.37009	$\{4,6,7,15\}$	$\{8,9,10,15\}$	0.203, 0.274, 0.459, 0.064
37	0.35279	$\{1,3,7,11,16\}$	$\{2,5,7,10,11\}$	0.255, 0.097, 0.169, 0.304, 0.174
41	0.34028	$\{1,3,9,13,17\}$	$\{3,5,10,13,14\}$	0.317, 0.187, 0.145, 0.166, 0.184

At $b = 2.5$ ($s = 1.2$) the maximiser is $S = F$ with equal shares for $p \leq 17$ (dense case), and sparse from $p = 19$ on ($|S| = 6, 8, 9, 10, 10, 12$ for $p = 19, 23, 29, 31, 37, 41$; data in `prog12.out`).

3.2 Linear gain models fail

Table 1 lists the free-hiding slopes. By Corollary 2 the game with gain $g \max_T R_T$ is won by the network iff $g > \min_T h_T$.

- With partner and riding only (the task as posed, all T alike): at $b = 2$, $h_1 < \frac{1}{2}$ for 29 of the 33 sparse features, minimum $h_1 = 0.095$ ($p = 41$, $l = 13$), and $h_1 < c_1 = 0.4226$ for 20 of them. At $b = 2.5$: $h_1 < \frac{1}{2}$ for all 55 sparse features, $\min h_1 = 0.124$ ($p = 29$, $l = 2$), and $h_1 < c_1 = 0.197$ for 10 of them. Dense cases: $h_1 = \frac{1}{2}$ exactly (Proposition 4), so $G = \frac{1}{2}R$ is exactly critical and $G = c_1 R$ is beaten by the factor $1/(2c_1) = 1.18$ ($b = 2$), 2.54 ($b = 2.5$).

- With stray outputs at $T = \{1, 2\}$: $h_{[2]} = 0$ for *every* sparse feature (33/38 at $b = 2$, 55/79 at $b = 2.5$), exactly as Proposition 3 predicts. So under any gain bound that is linear near $R = 0$ the network wins at every sparse prime. The protection at $T = [2]$ is then purely quadratic: $\Pi_{[2]}(R) \approx \kappa_2 R^2/a_l$ with $\min \kappa_2 = 0.82$ ($p = 13$, $l = 3$, $b = 2$) and 1.60 ($b = 2.5$).

Consequence for (O2): no proof can use a payment inequality that is linear in $|\beta_{l,T}|$ at small interference; the quadratic behaviour of the bonus near the IF block (Proposition 5) is necessary, not a convenience. Against $\kappa_2 \geq 0.82$ the local bonus constant on $T = [2]$ alone is $\frac{1-r}{8+r} = 0.0385$, a factor ≥ 21 of room at small R.

3.3 The sharp bonus

$G^*_{\mathrm{sym}}(\rho) = G^*(\rho, \rho, \rho)$ ($b = 2$): $G^*/\rho^2 = 0.350, 0.366, 0.392, 0.427$ at $\rho = 0.02, 0.1, 0.2, 0.3$ (local constant 0.346), maximum of G^*/ρ^2 equal to 0.687 at $\rho \approx 0.53$ (two-atom branch), G^*/ρ maximal in the interior (0.379 at $\rho_* = 0.572$, Proposition 6), then 0.331 at $\rho = 1$, rising to $c_1 = 0.4226$ at $\rho = \sqrt{3}$ (one atom). On $T = [2]$ alone: $G^*(0, \rho)/\rho^2 = 0.039 \to 0.05$ for $\rho \leq 0.3$ and $G^*(0, \rho) \leq (\frac{3}{2})^{1/2} - 1 = 0.2247$ for all ρ (attained from $\rho = 1.2247$, two aligned neurons). Spot checks with up to six atoms and 60 starts reproduce the two-atom values. The *critical configurations of the whole game are two-atom*: two equal atoms $c\, e^{\pm i\varphi/2}$ with

$$G_2(\rho_s, \rho_{12}) = (\tfrac{3}{2})^{s-1} \cos\big(\max(0, \arccos \tfrac{\rho_{12}}{2c} - 2 \arcsin \tfrac{\rho_s}{2c})\big) - 1, \qquad 2c = (\tfrac{3}{2})^{s-1},$$

which matches the computed G^* to 10^{-4} at the critical points ($G^*(0.545, 0.6) = 0.21335$, $\varphi/2 = 0.136$). In network terms: the cheap feature is realised by two neurons, slightly de-phased, and its three sign-flipped coefficients are all of order 0.5–0.7 a_l.

3.4 Verdicts with the sharp bonus

Table 1 ($b = 2$) lists for every feature: the original game of the task (partner + riding, prover's character at $\arg\max_T$, gain G^*_{sym}, fine ρ-grid), the T-resolved game with stray outputs and single-character Riesz tests (costs charged per test = relaxation in favour of the network; and with the true joint Lorentz account for the tightest cases), the quadratic test (L^2 over T, joint costs), and the single-T subgame ($\rho_s = 0$, quadratic test with $f = 1$ and Riesz).

Table 1: $b = 2$. h_1, h_{12}: free-hiding slopes; "lin": $h_1/\frac{1}{2}$, h_1/c_1 (< 1: network wins with linear gain); σ_{orig}: task's game; σ_{R}: T-resolved Riesz, separate costs; $\sigma_{\text{R}}^{\text{joint}}$: joint Lorentz account (tight cases); σ_{L^2}: quadratic test; last two: single-T subgame (quadratic, Riesz). $\sigma \geq 1 \Leftrightarrow V \geq 0$.

p	l	h_1	h_{12}	$\text{lin}_{1/2}$	lin_{c_1}	σ_{orig}	σ_{R}	$\sigma_{\text{R}}^{\text{joint}}$	σ_{L^2}	$\sigma_{\text{q}}^{[2]}$	$\sigma_{\text{R}}^{[2]}$
5	1	0.500	0.500	1.00	1.18	1.483	1.513	–	2.66	4.84	3.61
5	2	0.500	0.500	1.00	1.18	1.483	1.513	–	2.66	4.84	3.61
7	1	0.500	0.500	1.00	1.18	1.483	1.513	–	2.66	4.84	3.61
7	2	0.500	0.500	1.00	1.18	1.483	1.513	–	2.66	4.84	3.61
7	3	0.500	0.500	1.00	1.18	1.483	1.513	–	2.66	4.84	3.61
11	2	0.255	0.000	0.51	0.60	1.085	1.111	1.262	1.99	3.36	2.60
11	5	0.428	0.000	0.86	1.01	1.807	1.759	–	3.17	5.19	3.92
13	1	0.425	0.000	0.85	1.01	1.535	1.577	–	2.77	4.71	3.56
13	3	0.444	0.000	0.89	1.05	1.444	1.375	–	2.42	3.90	3.06
13	5	0.468	0.000	0.94	1.11	1.489	1.417	–	2.47	3.83	2.96
17	3	0.375	0.000	0.75	0.89	1.927	1.809	–	3.26	5.19	3.91
17	4	0.393	0.000	0.79	0.93	1.751	1.661	–	2.82	4.44	3.34
17	5	0.234	0.000	0.47	0.55	1.015	0.959	1.051	1.70	2.76	2.09
19	6	0.235	0.000	0.47	0.56	1.816	1.607	–	2.99	4.39	3.44
19	8	0.176	0.000	0.35	0.42	1.183	1.000	1.219	1.83	2.51	2.05
19	9	0.612	0.000	1.22	1.45	1.897	1.829	–	3.07	5.20	3.86
23	4	0.448	0.000	0.90	1.06	1.361	1.185	–	2.04	2.96	2.36
23	5	0.312	0.000	0.62	0.74	1.565	1.361	–	2.51	3.71	2.90
23	6	0.622	0.000	1.24	1.47	1.754	1.754	–	2.86	4.97	3.64
23	8	0.330	0.000	0.66	0.78	1.367	1.281	–	2.37	3.62	2.95
29	1	0.412	0.000	0.82	0.98	1.362	1.215	–	2.28	3.39	2.71
29	4	0.478	0.000	0.96	1.13	1.622	1.542	–	2.71	4.41	3.31
29	10	0.478	0.000	0.96	1.13	1.622	1.542	–	2.71	4.41	3.31
29	12	0.412	0.000	0.82	0.98	1.362	1.215	–	2.28	3.39	2.71
31	4	0.665	0.000	1.33	1.57	2.065	1.969	–	3.27	5.26	3.91
31	6	0.373	0.000	0.75	0.88	1.451	1.366	–	2.41	3.81	2.94
31	7	0.553	0.000	1.11	1.31	1.795	1.715	–	2.84	4.73	3.53
31	15	0.452	0.000	0.90	1.07	1.530	1.463	–	2.54	4.26	3.22
37	1	0.357	0.000	0.71	0.84	1.761	1.621	–	2.83	4.36	3.33
37	3	0.397	0.000	0.79	0.94	1.418	1.452	–	2.46	4.23	3.16
37	7	0.431	0.000	0.86	1.02	1.359	1.388	–	2.38	4.20	3.13
37	11	0.325	0.000	0.65	0.77	1.362	1.348	–	2.35	3.94	3.00
37	16	0.368	0.000	0.74	0.87	1.771	1.769	–	3.00	5.03	3.74
41	1	0.219	0.000	0.44	0.52	1.431	1.355	–	2.42	3.96	3.04
41	3	0.161	0.000	0.32	0.38	1.711	1.628	–	2.82	4.59	3.44
41	9	0.364	0.000	0.73	0.86	1.795	1.698	–	3.11	5.04	3.87
41	13	0.095	0.000	0.19	0.22	1.203	1.085	1.316	2.03	2.98	2.37
41	17	0.338	0.000	0.68	0.80	1.538	1.317	–	2.52	3.60	2.89

Grid caveat: σ_{R}, σ_{L^2} are minima over a grid of step 0.1 in the critical region (0.05–0.1 for L^2); local refinement with fresh G^* moves them by at most 3% ($p = 17$, $l = 5$: $\sigma_{\text{R}}^{\text{joint}}$ 1.078 $\to$ 1.051 at $(\rho_s, \rho_{12}) = (0.545, 0.6)$; σ_{L^2} 1.696 $\to$ 1.690 at $(0.45, 0.8)$). σ_{orig} is on a 120-point ρ-grid with $G = \max(G^*_{\text{sym}}, G_2)$.

Summary at $b = 2$.

- Task's game (partner + riding, sharp bonus): $\sigma_{\text{orig}} \geq 1.0146$; tightest $p = 17$, $l = 5$ (and its relabelled copies) at $\rho = 0.565$, then $p = 11$, $l = 2$ (1.084), $p = 19$, $l = 8$ (1.183), $p = 41$, $l = 13$ (1.203). So $V \geq 0$ in every case, but by 1.5% at $p = 17$.

- With stray outputs and single-character Riesz tests with *separate* costs the relaxed network wins: $\sigma_{\mathrm{R}} = 0.959$ ($p = 17$, $l = 5$, profile $(0.5, 0.6)$), 1.000 ($p = 19$, $l = 8$). Charging the payers once for all their modes (the true Lorentz account) restores the prover: $\sigma_{\mathrm{R}}^{\mathrm{joint}} = 1.051$ $(17, 5)$, 1.219 $(19, 8)$, 1.262 $(11, 2)$, 1.316 $(41, 13)$.

- The quadratic test is much stronger: $\sigma_{L^2} \geq 1.69$ for every feature ($p = 17$, $l = 5$), ≥ 1.83 for all others but one. The single-T subgame (the coordinator's "two aligned neurons" case) has $\sigma \geq 2.51$ (quadratic, $f = 1$) and ≥ 2.05 (Riesz).

Summary at $b = 2.5$. All margins are large: $\sigma_{\mathrm{orig}} \geq 2.42$ ($p = 23$, $l = 4$), $\sigma_{\mathrm{R}} \geq 2.35$ ($p = 41$, $l = 19$), $\sigma_{L^2} \geq 4.14$, single-$T \geq 4.55$. Linear gains still fail ($\min h_1/\frac{1}{2} = 0.25$, $\min h_1/c_1 = 0.63$; $h_{12} = 0$ for all sparse features). Full table: `tab12.tex`, `all12.txt`.

3.5 Variants

variant ($b = 2$)	$\min \sigma_{\mathrm{R}}$ (separate)	$\min \sigma_{L^2}$	worst feature
base (partner charged)	0.959	1.690	$p = 17$, $l = 5$
own partner free (note's (F1))	0.940	1.661	$p = 11$, $l = 2$ / $17, 5$
shared neurons (F2) free, any size	0.696	1.320	$p = 11$, $l = 2$ / $17, 5$

(F2) relations $2l \equiv \pm j \pm q$ ($j \neq q \in S$) exist for 30 of the 38 features at $p \leq 41$: at these sizes they are not rare. Treating them as free compensating vectors of unlimited size is a gross over-estimate (a shared neuron's cross mode is tied to its clean amplitudes and has its own siblings), and even then the quadratic test keeps $\sigma_{L^2} \geq 1.32$.

3.6 Dedicated compensators (with siblings)

Dedicated compensators (all factors unpaid) were modelled in two ways: (a) one neuron per frequency, cost $D(q) = 3^{1-s}\Gamma_s(1 - (1 - q)^s)$ with $q = \|C\|^r$, siblings left uncancelled and detected by their own single-character tests; (b) a sibling-free block, $q = (3^{s-1}\|C\|)^r$ (IF-block efficiency). Two searches at the critical profiles of the three tightest features ($p = 17, 19, 11$), with joint costs and the gain set to 0.98 of the threshold: (i) CCP (tangent majorisation of the concave cost) from random starts of sizes 0.005–0.4 a_l, all frequencies $k \in F$, both test characters: it always returns to $C = 0$; (ii) a pooled budget $\sum \|C_k\| \leq c$ charged $D((fc)^r)$ (a lower bound on the true concave cost): the value increases monotonically in $c \in [10^{-4}, 10^{-1}]a_l$ (e.g. $p = 17$: $0.0056 \to 0.0084$ ($c = 10^{-4}$) $\to 0.061$ (10^{-2}) $\to 0.25$ (10^{-1}) in model (b); model (a) similar), and beyond $0.1a_l$ the cost alone exceeds the largest possible bonus. *Dedicated compensators never help*: the concave cost $|C|^r$ is steepest exactly where the bonus is small.

3.7 Allocation moves (task 2)

Let the network move its allocation y on the budget sphere; the prover then uses *all* directions δ, with the exact coverage account $\mathrm{cov}_y(\mu) = 3^{1-s}(\Gamma_s \sum \mu - \sum_j y_j K_j(\mu))$ (this is the coverage gap $G(\theta)$ of the note minus what μ spends off the minimising directions). Results ($b = 2$):

	$p=17,\ l=5$	$p=19,\ l=8$	$p=11,\ l=2$	$p=41,\ l=13$
Riesz, separate: μ on B at $y^\star$	0.959	1.000	1.111	1.085
Riesz, separate: all δ at $y^\star$	1.133	1.366	1.474	1.622
Riesz, separate: $\mu=\mu^\star$ forced	0.766	0.655	1.051	0.740
Riesz, separate: min over y (NM, 80 evals)	0.999	1.120	1.462	1.182
Riesz, joint, at critical profile: $y^\star$ / min over y	1.151 / 1.095	1.447 / 1.341		
quadratic test: $y^\star$ / 4 moved y	2.004 / ≥ 1.995			

So: (i) the heuristic "the prover may use $\mu^\star$" is *not* sufficient: forcing $\mu=\mu^\star$ loses (0.655 at $p=19$). (ii) Allocation moves *do* help the network at first order (the value is a maximum of affine functions of dy whose active μ is not $\mu^\star$, so it decreases linearly in some directions), by up to 27% in σ_{R} and 5–7% in $\sigma_{\mathrm{R}}^{\mathrm{joint}}$; (iii) the decrease is bounded because moving y opens the coverage gap, which the prover collects with point masses; no allocation was found that brings $\sigma_{\mathrm{R}}^{\mathrm{joint}}$ or σ_{L^2} below 1.09 resp. 1.99. At $b=2.5$ ($p=41$, $l=19$ and $p=23$, $l=4$) local search found nothing below the value at $y^\star$ (2.43, 2.42). At small scales allocation moves are harmless (Section 2 gives the argument): if $R^2 \lesssim |dy|$ the prover's point mass at the worst direction collects a coverage gap $\gtrsim |dy|$ against a bonus $O(R^2)$; otherwise the $\mu^\star$-test margin, which is linear ($|T|=1$) or quadratic with constant $\kappa_2(\mu^\star)$ ($T=[2]$) in R, dominates the $O(|dy|^2)$ curvature of $\sum_j y_j K_j(\mu^\star)$. Numerically the $\mu^\star$-forced small-R constants are $\min h_1(\mu^\star)=0.077>0$ and $\min \kappa_2(\mu^\star)=0.298$, i.e. 7.7 times the local bonus constant 0.0385 on $T=[2]$ ($b=2$, all 38 features).

4 Theory: what can and cannot be proved, and the exact gap

$V \geq 0$ with a linear gain is false. With $G=\frac{1}{2}R$ or $G=c_1 R$ the statement "$V\geq 0$ for all p" fails already in the model with partner and riding only (Corollary 2: $h_1 < c_1$ for 20 sparse features at $b=2$), and it fails for *every* sparse feature once stray outputs are admitted (Proposition 3(b): $h_{[2]}=0$). The obstruction is exact and elementary: near-hiding of the sign-flipped mode at quadratic cost. It is decided by a count (Chebotarev): free first-order hiding at $\chi_{l,T}$ is possible iff the channels give $\geq |B|$ distinct compensating output frequencies. This also corrects the reading of Proposition *hide*: exact hiding with the features' own outputs is impossible only for partner + riding ($2|S|-1<|B|$); stray outputs on $T=[n]$ add up to $2|S|$ more and make it possible. The dense case is the extreme opposite: there the slope is exactly $\frac{1}{2}$, so Theorem A's constant is *exactly* the threshold.

The quadratic bonus is necessary and (near IF) proved. Proposition 5 gives the exact local constant. The suggested combination $G\leq \min(\frac{1}{2}R,\, CR^2/a)$ with the best global C ($C=0.687$ at $b=2$, attained on the two-atom branch at $\rho\approx 0.53$) is *not* enough: against it the protection ratio drops to 0.81 (partner + riding, $p=17$, $l=5$; 0.93 with c_1 in place of $\frac{1}{2}$) and to 0.57 once stray outputs are allowed ($p=19$, $l=8$). What works numerically is the *T-resolved* sharp bonus $G^*(\rho_s,\rho_{12})$, whose decisive features are (i) the small constant $\frac{1-r}{8+r}$ when the interference sits on $T=[2]$, where hiding is exact, and (ii) the explicit two-atom branch G_2 in the critical region.

Where the game is decided. Two regimes, as anticipated:

(S) *Small R.* On $|T|=1$ the protection is linear ($h_1\geq 0.095$) against a quadratic bonus; on $T=[2]$ both are quadratic, with protection constant $\kappa_2\geq 0.82$ against bonus constant 0.0385 ($b=2$). Cross terms between the two are cubic. This regime is safe by a factor ≥ 20,

uniformly in the cases computed, and robust under allocation moves ($\mu^\star$-forced constants 0.077, 0.298).

(L) *Moderate R, two-atom configurations* ($\rho_s \approx 0.5$, $\rho_{12} \approx 0.6$–0.8, bonus $\approx 0.21\, a_l$). Here the Lorentz costs and the residual are of the same order as the bonus. This is where all the tight cases sit: $\sigma_{\text{orig}} = 1.015$, $\sigma_{\text{R}}^{\text{joint}} = 1.051$, $\sigma_{L^2} = 1.69$ at $p = 17$, $l = 5$; the separate-cost relaxation (0.959) and the forced-$\mu^\star$ test (0.655 at $p = 19$) lose here.

(X) *Large R* ($\rho \to \sqrt{3}$, one atom): compensation becomes expensive ($|z| \to a_j$) and every ratio is comfortable.

A sufficient, checkable form of (O2) for one cheap feature at $n = 2$. Combining Lemma 1 with the quadratic test, the following two inequalities imply that a single cheap feature (with any partner, riding and stray compensation, dedicated compensators and allocation moves included as in Section 3) never beats IF at the programme vertex (S, B):

($\text{P}^\sharp$) *payment*: for every atom system, $|A| - a \le a\, G^\sharp\big(\max(|\beta_1|, |\beta_2|)/a,\ |\beta_{12}|/a\big)$, for an explicit $G^\sharp$ that is $\ge G^*$ (e.g. $G^\sharp = \max(G_2,\ \frac{1-r}{8+r}(2\rho_s + \rho_{12})^2 + \text{cubic})$ near 0, $c_1 \max \rho$ for large ρ);

($\text{A}^\sharp$) *protection*: $\Pi^{L^2}(\rho_s, \rho_s, \rho_{12}) \ge a_l\, G^\sharp(\rho_s, \rho_{12})$ on $[0, 3^{s-1}]^2$, where Π^{L^2} is the SOCP value of Section 1 (quadratic test, joint Lorentz costs).

($\text{A}^\sharp$) holds numerically with factor ≥ 1.69 ($b = 2$) and ≥ 4.1 ($b = 2.5$) for every feature at $p \le 41$, with the computed G^* in place of $G^\sharp$. ($\text{P}^\sharp$) is a single-feature inequality on atoms (the paper's Lemma 6 makes it network-independent); only its linear envelope (Theorem A) and its local quadratic form (Proposition 5) are proved.

The precise remaining gap for (O2).

1. *($P^\sharp$), a T-resolved sharp payment at $n = 2$.* Needed: a proof that the bonus is at most (a) $\frac{1-r}{8+r}|\beta_1 + \beta_2 + \overline{\beta_{12}}|^2/a$ plus higher order *globally on a neighbourhood of size independent of* p, and (b) at most the two-atom value G_2 in the critical region $\rho \in [0.4, 0.8]$, with (c) the known linear cap. The numerics say there is room ($\ge 69\%$ with the quadratic test), so a non-sharp version with explicit constants suffices; for instance any $G^\sharp \le 1.6\, G^*$ would do for all $p \le 41$ at $b = 2$. The spectral form of the cost lemma (main thread) and the chain-matrix second-order expansion are the natural tools; the two-atom branch is the extremal family to beat.

2. *($A^\sharp$) uniformly in p.* The protection curves are explicit SOCP values attached to (S, B, μ) but their constants vary irregularly with the arithmetic of the support ($h_1 \in [0.095, 0.67]$, $\kappa_2 \in [0.8, 40]$ at $b = 2$). A proof for all p needs a *robust* uncertainty principle on $\mathbb{Z}_p$: a lower bound for the weighted $L^1(\mu)$ distance from $c_l|_B$ to the span of $\le |B| - 1$ characters (for $|T| = 1$), and for the Lorentz-weighted quadratic form when the span is everything (for $T = [2]$). The dichotomy (Proposition 3) is the exact, non-quantitative version.

3. *Several cheap features.* The game here has one cheap feature whose compensation is paid by IF-priced payers. With several cheap features the payers' own bonuses and the (F2)/(O1) relations interact; the quadratic test handles the relations only under (D_3)-type dissociation.

What is *numerical* in this note: all σ values, the sharp bonus G^* (local search; a lower bound for the true supremum, so every σ is an upper estimate of the true ratio, up to grid resolution $\leq 3\%$), the dedicated and allocation searches (local). What is *proved*: Lemma 1, Corollary 2, Propositions 3, 4, 5 (local) and 6.

Files (scratchpad `fork02/`)

`progdata.py` (programme, binding set, KKT measure; `prog15.out`, `prog12.out`), `game2.py` (dual SOCP, channel families, free-hiding slope), `gameL2.py` (quadratic test), `fb.py`, `runb.py`, `grid2d.py` (sharp bonus G^*; `grid_s*.json`, `G_*_s*.json`), `check2nd.py` (Proposition 5), `curves2.py`, `curvesL2.py` (protection curves), `sigma_all.py`, `sigma_orig.py`, `refine_orig.py`, `jointsig.py`, `refine_joint.py`, `refineL2.py` (verdicts; `all15.txt`, `all12.txt`, `refine15.txt`, `refine12.txt`), `ded.py`, `dedpool.py`, `dedjoint.py` (dedicated compensators), `alloc.py`, `allocsearch.py`, `allocjopt.py`, `allocL2y.py`, `musig.py`, `musmall.py` (allocation moves).